

Module 5: Transport Layer & Security

TCP 3-Way Handshake, TCP AIMD Congestion Control, DNS Hierarchy, HTTP Evolution & RSA Public-Key Math

Module V: Ethernet, IP, Routing & Applications — Complete 16-Topic Tracker

- Topic 39:** Traditional Ethernet (IEEE 802.3 CSMA/CD at 10 Mbps)

Topic 41: IEEE 802.11 Wi-Fi (CSMA/CA with RTS/CTS, Ethernet vs. Wi-Fi)

Topic 43: IP Addressing & Subnetting (2^{borrowed} & $2^{\text{host}} - 2$)

Topic 45: Routing in Packet Networks (Hop Count, Cost, Bandwidth)

Topic 47: Link State Routing (Dijkstra's Algorithm & OSPF Flooding)

Topic 49: Congestion Control (Leaky Bucket vs. Token Bucket Shaping)

Topic 51: Simple Mail Transfer Protocol (SMTP, POP3, IMAP)

Topic 53: Hypertext Transfer Protocol (HTTP/1.1 vs. HTTP/2 & Statuses)
- Topic 40:** High-Speed Ethernet (Fast 100M, Gigabit 1G, 10-Gigabit)

Topic 42: Internet Protocol (IPv4 20-Byte Header vs. IPv6 128-Bit)

Topic 44: Transport Protocols (TCP 3-Way Handshake vs. UDP Sockets)

Topic 46: Distance Vector Routing (Bellman-Ford & RIP Count-to-Infinity)

Topic 48: Path Vector Routing (Border Gateway Protocol BGP-4)

Topic 50: Traffic Management & Quality of Service (QoS Dimensions)

Topic 52: Domain Name System (DNS Hierarchy & Records A, MX)

Topic 54: Dynamic Host Configuration Protocol (DHCP DORA Sequence)

Topic 39 – 41: Ethernet Evolution & Wireless LAN (IEEE 802.11)

Standard	Data Rate	Physical Media	Medium Access Control (MAC)
Traditional Ethernet	10 Mbps	10Base5 (Thick coax), 10BaseT (UTP)	CSMA/CD (Half-Duplex)
Fast Ethernet	100 Mbps	100BaseTX (Cat 5 UTP), 100BaseFX	CSMA/CD / Switched Full-Duplex
Gigabit Ethernet	1 Gbps	1000BaseT (Cat 5e UTP), 1000BaseLX	Switched Full-Duplex
IEEE 802.11 (Wi-Fi)	Up to 9.6 Gbps (Wi-Fi 6)	2.4 GHz/5 GHz/6 GHz Radio	CSMA/CA with RTS/CTS Handshake

Topic 42 & 43: Internet Protocol (IPv4 Header & Subnetting) [UPLOADED PYQ]

IPv4 Header Field [UPLOADED PYQ]	Bit Width	Functional Operational Purpose
Version	4 bits	Identifies IP version (`0100` for IPv4).
Header Length (IHL)	4 bits	Header length in 32-bit words (Minimum value = 5 $\implies$ 20 bytes).
Total Length	16 bits	Total datagram length (Header + Payload) in bytes (Max = 65,535 bytes).
Identification, Flags, Offset	16 + 3 + 13 bits	Handles IP fragmentation. `DF`=Don't Fragment, `MF`=More Fragments. Offset in 8-byte units.
Time to Live (TTL)	8 bits	Decrement at each router hop. When TTL = 0, packet is dropped (prevents infinite loops).
Protocol	8 bits	Demultiplexes to Transport layer (`6` for TCP, `17` for UDP, `1` for ICMP).
Header Checksum	16 bits	1's complement sum of header fields recalculated at each router hop.

Subnetting Formulations: - Number of Created Subnets = $2^{\text{borrowed bits}}$ - Total IP Addresses per Subnet = $2^{\text{host bits}}$ - Usable Host Addresses per Subnet = $2^{\text{host bits}} - 2$ (subtracting Subnet Network ID and Directed Broadcast Address).

Topic 44: Transport Protocols (TCP vs. UDP) [UPLOADED PYQ]

Feature	TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
Connection Model	Connection-Oriented (3-Way Handshake: SYN $\rightarrow$ SYN-ACK $\rightarrow$ ACK).	Connectionless (no handshake; sends datagrams immediately).
Reliability	100% Reliable (Sequence numbers, ACKs, retransmissions on loss).	Unreliable Best-Effort (no ACKs, lost packets are not retransmitted).
Ordering & Streaming	Guaranteed strictly in-order byte stream reassembly.	No ordering guarantee; datagram boundary preserved.
Flow & Congestion Control	Full sliding window flow control + AIMD Congestion Control.	Zero flow control, zero congestion control.
Typical Applications	Web (HTTP/HTTPS), Email (SMTP), File Transfer (FTP), SSH.	DNS queries, Video streaming (RTP), VoIP, DHCP, Online Gaming.

Topic 45 – 48: Routing Protocols (Distance Vector vs. Link State vs. Path Vector)

Routing Protocol	Mathematical Algorithm	Scope & Protocol	Key Characteristics
Distance Vector	Bellman-Ford: $D_x(y) = \min_v \{c(x, v) + D_v(y)\}$	Intra-AS (RIP)	Exchanges vector with neighbors; susceptible to Count-to-Infinity.
Link State	Dijkstra's Shortest Path Tree Algorithm	Intra-AS (OSPF)	Floods Link State Advertisements (LSA) globally; builds complete topological map.
Path Vector	Policy-based sequence of Autonomous System (AS) hops	Inter-AS (BGP-4)	Prevents loops by checking if local AS number is present in path vector.

Topic 49 & 50: Congestion Control & Traffic Shaping (Leaky vs. Token Bucket)

Traffic Shaper	Operating Mechanism	Burst Handling Capability
1. Leaky Bucket	Input packet bursts fill a buffer with fixed capacity. Water (packets) leaks out at a strictly constant uniform output rate . Excess bursts overflow and are discarded.	Completely eliminates bursts; outputs strictly smoothed constant bit-rate traffic.
2. Token Bucket	Tokens generate into the bucket at a constant rate r up to capacity b . A packet can only transmit if it captures tokens.	Allows controlled, regulated bursts up to bucket capacity b while capping average rate at r .

Topic 51 – 54: Application Layer Services (SMTP, DNS, HTTP, DHCP)

Summary of Core Application Protocols & Ports

- **SMTP (Port 25/587)**: Simple Mail Transfer Protocol; pushes email between client and server and between mail relays.
- **DNS (Port 53 – UDP/TCP)**: Hierarchical Domain Name System resolving domain names to IP addresses (Records: `A` for IPv4, `AAAA` for IPv6, `MX` for Mail Exchange, `CNAME` for Canonical Alias).
- **HTTP/HTTPS (Port 80/443)**: Stateless request-response web protocol. Status codes: `200 OK`, `301 Moved Permanently`, `404 Not Found`, `500 Internal Server Error`.
- **DHCP (Port 67/68 – UDP)**: Dynamic Host Configuration Protocol assigning IP addresses dynamically via the **D-O-R-A** sequence (Discover → Offer → Request → Acknowledge).

[UPLOADED PYQ] Complete Life Cycle: What Happens When You Type `https://example.com`?

1. **Host IP Configuration (DHCP):** Client uses DHCP DORA sequence to acquire local IP, Subnet Mask, Default Gateway, and DNS Server.
2. **DNS Resolution:** Browser queries local DNS cache, then queries Recursive Resolver → Root Server → TLD Server (`.com`) → Authoritative Server to obtain target IP for `example.com`.
3. **ARP Translation:** Host uses Address Resolution Protocol (ARP) to map Default Gateway IP to its physical 48-bit MAC address.
4. **TCP 3-Way Handshake:** Client sends `SYN` (Port 443) → Server replies `SYN-ACK` → Client confirms `ACK` (Connection Established).
5. **TLS 1.3 Handshake:** Client and server negotiate cipher suites, authenticate server X.509 certificate, and exchange ephemeral Diffie-Hellman keys to establish encrypted session.
6. **HTTP GET Request & IP Routing:** Browser transmits encrypted `GET / HTTP/2` request. Packet is encapsulated with IP header and routed across intermediate Autonomous Systems via OSPF and BGP-4.
7. **Server Processing & Response:** Web server processes request and returns HTTP `200 OK` response with HTML/CSS payload.
8. **Browser Rendering:** Browser parses HTML DOM tree, fetches CSS/JS, and renders the visual webpage!

M5 Active Recall & Exam Questions [UPLOADED PYQ]

Q1. [UPLOADED PYQ] Explain the structure and working of the IPv4 Header with the purpose of all major fields. (10 Marks)

The IPv4 header has a minimum size of 20 bytes (up to 60 bytes with options):

1. *Version (4b):* `4` for IPv4.
2. *IHL (4b):* Header length in 32-bit words (min 5).
3. *Total Length (16b):* Header + Data in bytes.
4. *Identification (16b), Flags (3b), Fragment Offset (13b):* Fragmentation management for MTU limits.
5. *TTL (8b):* Hop limit decremented at each router to prevent packet looping.
6. *Protocol (8b):* Transport demultiplexing (`6`=TCP, `17`=UDP).
7. *Header Checksum (16b):* Error detection across header fields.
8. *Source IP (32b) & Destination IP (32b):* Logical sender and receiver addresses.